Multiple myeloma & precursors

MGUS → smoldering → myeloma · **treat on a myeloma-defining event, not on the M-spike** · anti-CD38 quadruplets up front, T-cell redirection (CAR-T, bispecifics) moving to first relapse

Compiled from primary trial reports and consensus guidelines (IMWG; NCCN) verified against PubMed, September 2026.

THE PLASMA CELL SPECTRUM (IMWG 2014)

Entity	Definition
MGUS *~3% of adults >50*	M-protein **<3 g/dL** + clonal marrow PCs **<10%** + **no CRAB** / **SLiM** (or amyloidosis). Progresses **~1%/yr**; IgM MGUS → WM, not myeloma
Light-chain MGUS	Abnormal FLC ratio + raised involved FLC, **no heavy chain on IFE**, PCs <10%, urine M <500 mg/24 h
Smoldering (SMM)	M-protein **≥3 g/dL** (or urine ≥500 mg/24 h) **and/or PCs 10–60%**, no CRAB / SLiM. **~10%/yr for the first 5 yr**, then ~3%/yr, then ~1%/yr
Multiple myeloma	Clonal PCs **≥10%** (or biopsy-proven plasmacytoma) **+ ≥1 myeloma-defining event** (below)
Plasma cell leukemia	**≥5% circulating PCs** (IMWG 2021; was 20%) — treat as ultra-high risk
Solitary plasmacytoma	Single lesion, marrow <10% clonal (none = solitary; minimal = higher risk). **RT 40–50 Gy**

MYELOMA-DEFINING EVENTS — ANY ONE = TREAT

- **C** — Ca **>11 mg/dL** or >1 mg/dL above ULN · **R** — CrCl **<40** or Cr **>2 mg/dL** · **A** — Hb **<10 g/dL** or >2 below LLN · **B** — **≥1 lytic lesion** (≥5 mm) on X-ray, CT or PET-CT
- **SLiM** (no CRAB needed): clonal PCs **≥60%** · involved:uninvolved **FLC ratio ≥100** with involved FLC ≥100 mg/L · **>1 focal lesion on MRI** (≥5 mm)
- Osteoporosis / compression fracture **without a lytic lesion** is not CRAB — image first

WORKUP

- **SPEP + serum IFE + serum FLC** — plus **24-h UPEP + urine IFE** once at diagnosis
- CBC, CMP (Ca, Cr, albumin), **β2-microglobulin, LDH**, quantitative IgG/IgA/IgM
- **Marrow aspirate + biopsy**: PC %, flow, **FISH on CD138-sorted cells** — t(4;14), t(14;16), t(14;20), t(11;14), del(17p), 1q+, del(1p); **TP53** sequencing where available
- **Whole-body low-dose CT** or **PET-CT**; **whole-body MRI** if SMM with a negative CT/PET. **Skeletal survey is obsolete** (insensitive)
- Albuminuria or unexplained organ dysfunction → think **AL amyloidosis** (fat pad, NT-proBNP, troponin)
- **Type & screen with extended RBC phenotype/genotype before the first anti-CD38 dose** (page 3)

TREATMENT PATHWAY

Smoldering myeloma

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Risk-stratify (Mayo 2/20/20)
low/intermediate: **observe**, labs q3–6 mo
high risk: daratumumab SC ×3 yr (AQUILA)
or lenalidomide (E3A06), or a trial

Newly diagnosed — transplant-eligible

↓

Quadruplet induction ×4
D-VRd (PERSEUS) · collect stem cells early
alternatives: Isa-VRd, D-KRd (high risk)

↓

Melphalan 200 → ASCT → consolidation
ASCT: **PFS gain, no OS gain** (DETERMINATION)
MRD-neg after quad: deferral OK (MIDAS)

↓

Maintenance — D-R
dara q4wk + len 10 mg (PERSEUS)
stop dara at ≥24 mo if CR + MRD-neg ≥12 mo
len alone if no dara induction / not tolerated

Newly diagnosed — transplant-ineligible

↓

Fit / intermediate (IMWG frailty)
D-VRd (CEPHEUS) or **Isa-VRd** (IMROZ)
or **DRd** (MAIA, OS benefit) until progression

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Frail
DRd with dex 20 mg weekly
or **VRd-lite**; avoid 4-drug intensity

Relapse

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First relapse — usually len-refractory
switch classes; **cilta-cel** (OS benefit)
tec + dara (OS benefit) · BVd · anti-CD38 + K/P

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Triple-class exposed and beyond
BCMA CAR-T or **BCMA bispecific**
talquetamab (GPRC5D) after BCMA
selinexor · **venetoclax only if t(11;14)**

SMOLDERING MYELOMA — TREATMENT

- **Mayo 2/20/20:** M-protein **>2 g/dL**, marrow PCs **>20%**, FLC ratio **>20**. 0 / 1 / ≥2 factors → median time to progression **110** / **68** / **29 months**
- **High-risk SMM → daratumumab SC ×3 yr** (AQUILA vs active monitoring): PFS HR **0.49**, 5-yr PFS 63% vs 41%, **OS HR 0.52** — **FDA 6 Nov 2025**, the first approved SMM therapy
- **Lenalidomide** (E3A06, intermediate/high risk): PFS HR 0.28; no OS signal
- **Low-risk SMM: observe** — treating it is not supported

MGUS — MAYO RISK (20-YR PROGRESSION)

Risk factors	20-yr risk
M-protein ≥1.5 g/dL · non-IgG · abnormal FLC ratio — 0 factors	**5%**
1 · 2 · 3 factors	21% · 37% · **58%**

Low-risk MGUS (IgG, M <1.5, normal ratio): **no marrow or imaging needed**; recheck at 6 mo then every 2–3 yr. Anything else: marrow + imaging, then yearly labs.

STAGING

System	Definition · outcome
ISS	I: β2M **<3.5** mg/L + albumin **≥3.5** g/dL · III: β2M **≥5.5** · II: neither
R-ISS (2015)	I: ISS I + standard-risk FISH + normal LDH · **III: ISS III + high-risk FISH [del(17p), t(4;14), t(14;16)] or high LDH** · II: rest. 5-yr OS **82** / **62** / **40%**
R2-ISS (2022)	Points: ISS III 1.5, ISS II 1, del(17p) 1, high LDH 1, t(4;14) 1, **1q+ 0.5** → I 0 · II 0.5–1 · III 1.5–2.5 · IV 3–5. Median OS NR / 109 / 69 / 38 mo. **Splits the big R-ISS II group**

HIGH-RISK MYELOMA — IMS/IMWG 2025 (ANY ONE)

- **del(17p) in >20% cancer clonal fraction and/or TP53 mutation**
- **IgH translocation [t(4;14), t(14;16), t(14;20)] + 1q gain/amp and/or del(1p32)** — translocation **alone no longer qualifies**
- **Biallelic del(1p32)**, or **monoallelic del(1p32) + 1q gain/amp**
- **β2M ≥5.5 mg/L with normal creatinine** (<1.2 mg/dL)
- Also high risk in practice: **plasma cell leukemia**, extramedullary disease, **relapse <12–18 mo** from front-line therapy (functional). **t(11;14), t(6;14), hyperdiploidy = standard risk**

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NEWLY DIAGNOSED — KEY TRIALS

Trial	Design · result
PERSEUS *TE*	**D-VRd vs VRd** induction/consolidation + ASCT, then **D-R vs R**: 6-yr PFS **81.1% vs 56.1%** (HR 0.35, 81-mo update, meeting report); MRD-neg **75% vs 48%.** Dara stopped after ≥24 mo if sustained MRD-neg. FDA 30 Jul 2024 for **D-VRd induction + consolidation only** (not D-R maintenance)
DETERMINATION · IFM 2009	RVd ± ASCT, then len: PFS **67.5 vs 46.2 mo** (IFM: 50 vs 36) — **no OS difference** (5-yr 80.7% vs 79.2%). **ASCT buys PFS, not OS**; deferral with cells banked is legitimate
MIDAS *IFM 2020-02*	Isa-KRd ×6, then MRD-guided. **MRD-neg (10^{-5}): ASCT vs 6 more Isa-KRd** — MRD-neg 10^{-6} 86% vs 84% (P=0.64). MRD+: tandem vs single ASCT 32% vs 40% (NS). **MRD endpoint only**, 17-mo follow-up, no PFS yet
Len maintenance *meta-analysis*	After ASCT: PFS **52.8 vs 23.5 mo**, **OS HR 0.75**; more second primary malignancies
AURIGA	MRD+ after ASCT: **D-R vs R maintenance** — MRD-neg conversion at 12 mo **50.5% vs 18.8%**; PFS HR 0.53
MAIA *TIE*	**DRd vs Rd** until progression: median PFS NR vs 34.4 mo (HR 0.53), **OS HR 0.68**
SWOG S0777	**VRd vs Rd**: PFS 43 vs 30 mo, **OS 75 vs 64 mo** (HR 0.71)
CEPHEUS *TIE / deferred*	**D-VRd vs VRd** ×8 → D-Rd vs Rd: MRD-neg **60.9% vs 39.4%**; PFS HR **0.57**. FDA 27 Jan 2026 (TIE only)
IMROZ *TIE, ≤80 yr*	**Isa-VRd vs VRd**: 5-yr PFS **63.2% vs 45.2%** (HR 0.60). FDA Sep 2024

No head-to-head quadruplet vs DRd in TIE. Quadruplets for fit older adults; **DRd for intermediate-fit / frail**.

FITNESS AND DOSE ADJUSTMENT

- **IMWG frailty score** (age, Charlson, ADL, IADL; 0–5): **fit 0 · intermediate 1 · frail ≥2** — 3-yr OS 84% / 76% / 57%; frail have ~2× the discontinuation rate
- Age >75 or frail: **dexamethasone 20 mg weekly**, **weekly SC bortezomib**, renally dosed lenalidomide

RESPONSE (IMWG 2016)

Level	Criteria
MRD-neg	CR + no clonal PCs at **≤10^{-5}** (NGF or NGS); **sustained** = confirmed ≥1 yr apart; imaging MRD-neg adds PET-negative
sCR / CR	**Negative serum + urine IFE**, plasmacytomas gone, **<5% marrow PCs** (sCR adds normal FLC ratio, no clonal PCs by IHC)
VGPR / PR	≥90% M-protein fall (or IFE+ / SPEP–) / ≥50% serum and ≥90% urine fall
Progression	≥25% rise from nadir: serum M **≥0.5 g/dL absolute**, urine ≥200 mg/24 h, or dFLC >10 mg/dL; new lesions

MRD is prognostic, and now regulatory: ODAC 12–0 (Apr 2024) accepted MRD-neg CR as an accelerated-approval endpoint. CEPHEUS (Jan 2026) used MRD-neg as its primary endpoint; **iberdomide (Aug 2026) = first accelerated approval of a new drug based on MRD**. **Stopping therapy on MRD alone is still a trial question.**

RELAPSE — BY REFRACTORINESS

Setting	Options · evidence
Definitions	**Triple-class exposed** = PI + IMiD + anti-CD38. **Refractory** = progression on or ≤60 days after. **Penta** = 2 PIs + 2 IMiDs + anti-CD38. Treat biochemical relapse early if high-risk or rapid rise
Len-refractory, 1–3 lines *the usual first relapse*	**Cilta-cel** (CARTITUDE-4): PFS NR vs 11.8 mo (HR 0.26), **OS HR 0.55**. **Teclistamab + dara** (MajesTEC-3): 36-mo PFS **83% vs 30%** (HR 0.17), **OS HR 0.46**. **BVd** (DREAMM-7): PFS 36.6 vs 13.4 mo, **OS HR 0.58**
Anti-CD38-naive, 1–3 lines	**Dara-Kd** (CANDOR, HR 0.63) · **Isa-Kd** (IKEMA, HR 0.53) · **DPd** (APOLLO, 12.4 vs 6.9 mo) · **Isa-Pd** (ICARIA, 11.5 vs 6.5 mo) · **IberDd** (iberdomide, **Zenbexus**; accelerated Aug 2026, ≥1 prior line incl PI + IMiD)
Anti-CD38 + len-refractory	**Teclistamab monotherapy** (MajesTEC-9 vs PVd/Kd): PFS HR **0.29**, **OS HR 0.60**, grade 3–4 infection 42%. **BPd** (DREAMM-8 vs PVd): 12-mo PFS 71% vs 51%. CAR-T
≥4 lines, triple-class	BCMA CAR-T, BCMA or GPRC5D bispecific (table below) · **SVd** (BOSTON: PFS 13.9 vs 9.5 mo) · clinical trial

Cilta-cel vs ide-cel: ide-cel (KarMMa-3, 2–4 prior lines) PFS 13.3 vs 4.4 mo but **no OS gain** (HR 1.01, 56% crossover). **Both CAR-Ts show early excess deaths** during manufacturing/bridging — bridge effectively.

T-CELL REDIRECTION AND ADC — APPROVALS

Agent	Target · label · key data
Cilta-cel (CAR-T)	BCMA · **≥1 prior line, len-refractory** · **OS benefit**. **Delayed parkinsonism**, cranial-nerve palsy (9%), GBS
Ide-cel (CAR-T)	BCMA · ≥2 prior lines (IMiD, PI, anti-CD38) · PFS benefit, OS not shown
Teclistamab	BCMA×CD3 SC · **+ dara, ≥1 line** (5 Mar 2026) · mono ≥4 lines (ORR 63%) **converted to regular approval 5 Mar 2026**
Elranatamab	BCMA×CD3 SC · ≥4 lines (accelerated; ORR 61%)
Linvoseltamab	BCMA×CD3 IV · ≥4 lines (accelerated Jul 2025; ORR 71%, ≥CR 50%)
Talquetamab	**GPRC5D**×CD3 SC · ≥4 lines (accelerated; ORR ~70%) · **works after BCMA**; oral/skin/nail toxicity, fewer infections
Belantamab (ADC)	BCMA · **BVd, ≥2 lines incl PI + IMiD** (re-approved Oct 2025; withdrawn 2022 after DREAMM-3) · **keratopathy**, REMS

Sequencing: BCMA loss after BCMA therapy is common — after BCMA CAR-T/bispecific, **switch target (GPRC5D)** rather than re-using BCMA when possible.

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REGIMEN DOSING

Regimen	Dosing
VRd *21-day*	Bortezomib **1.3 mg/m² SC d1, 4, 8, 11** + lenalidomide **25 mg d1–14** + dex 20 mg d1, 2, 4, 5, 8, 9, 11, 12 (S0777, ×8)
D-VRd *28-day (PERSEUS)*	**Dara SC 1800 mg** weekly c1–2, q2wk c3–6 + V 1.3 d1, 4, 8, 11 + **R 25 d1–21** + dex 40 d1–4, 9–12 ×4 → ASCT → ×2 → **D-R** (dara q4wk)
VRd-lite *35-day*	Bortezomib 1.3 SC **weekly** d1, 8, 15, 22 + len 15 d1–21 + dex 20 twice weekly
DRd *28-day (MAIA)*	Dara SC **1800 mg: weekly wk 1–8, q2wk wk 9–24, q4wk from wk 25** + len 25 d1–21 + dex 40 weekly (20 if >75)
Isa-VRd (IMROZ)	Isatuximab **10 mg/kg IV** weekly ×4, then q2wk; later q4wk + VRd ×4 → Isa-Rd
KRd *28-day*	Carfilzomib **20 → 27 mg/m² d1, 2, 8, 9, 15, 16** (or 56–70 once weekly) + len 25 d1–21 + dex 40 weekly
Dara-Kd / Isa-Kd	Carfilzomib **20 → 56 mg/m²** d1, 2, 8, 9, 15, 16 + dex 40 weekly + anti-CD38
DPd / Isa-Pd	Pomalidomide **4 mg d1–21** q28d + dex 40 weekly (20 if ≥75) + dara SC or isatuximab 10 mg/kg
BVd	Belantamab **2.5 mg/kg IV q3wk** + V 1.3 d1, 4, 8, 11 + dex 20 ×8 cycles, then belantamab alone. Eye-toxicity reductions: 1.9 mg/kg q3wk → 1.9 q8wk
SVd	Selinexor **100 mg PO once weekly** + bortezomib 1.3 SC weekly + dex 20 twice weekly (35-day)
IberDd (Zenbexus)	Iberdomide **1 mg d1–21** q28d + dara SC + dex 40 weekly
Dara SC (AQUILA, SMM)	**1800 mg** weekly c1–2, q2wk c3–6, then q4wk — **stop at 36 mo** (39 cycles)
Len maintenance	**10 mg d1–28** (15 after 3 mo if tolerated); CrCl 30–60 → 5 mg; <30 → 2.5 mg; dialysis 5 mg after HD
Mel 200 + ASCT	Melphalan **200 mg/m²** day −2; **140 mg/m²** if renal impairment or frail (100–140 on dialysis); oral cryotherapy
Teclistamab SC	**0.06 → 0.3 mg/kg** step-up, then **1.5 mg/kg weekly**; q2wk after ≥CR ≥6 mo. **With dara:** 1.5 weekly wk 2–8 → **3 mg/kg q2wk** wk 9–24 → q4wk
Elranatamab SC	**12 mg d1 → 32 mg d4 → 76 mg** weekly; q2wk from wk 25 if responding, q4wk from wk 49
Talquetamab SC	0.01 → 0.06 mg/kg → **0.4 mg/kg weekly**, or + 0.4 step → **0.8 mg/kg q2wk**
Linvoseltamab IV	**5 → 25 → 200 mg** (d1, 8, 15), 200 weekly wk 4–13, **q2wk from wk 14**; q4wk from wk 24 if ≥VGPR
CAR-T	Flu 30 + Cy 300 mg/m² ×3 d. **Cilta-cel 0.5–1.0×10^6 CAR+ cells/kg** · **ide-cel 300–510×10^6**. Stay near center 2 wk, **no driving 2 wk** (REMS removed Jun 2025)

Lenalidomide renal dosing (induction): CrCl 30–60 → 10 mg daily · <30 → 15 mg q48h · dialysis → 5 mg daily after HD. **Bortezomib, daratumumab, carfilzomib, pomalidomide (off HD) need no renal adjustment**; bortezomib: reduce for hepatic impairment.

SUPPORTIVE CARE

- **Bone: zoledronic acid 4 mg IV q4wk** (q12wk after ~1–2 yr) or **denosumab 120 mg SC q4wk** for every patient on active therapy (non-inferior, HR 0.98; **less renal toxicity** 10% vs 17%, more hypocalcemia 17% vs 12%). **Denosumab if CrCl <30**; don't stop it abruptly (rebound fractures). Zoledronic acid renal dosing: CrCl 50–60 → 3.5 mg · 40–49 → 3.3 · 30–39 → 3.0 · <30 → avoid
- **ONJ:** dental evaluation before starting; avoid extractions on therapy. **Calcium + vitamin D** for all (not if hypercalcemic)
- **VTE with any IMiD: aspirin 81–325 mg** if ≤1 risk factor; **prophylactic LMWH or apixaban 2.5 mg BID / rivaroxaban 10 mg** if higher risk (IMPEDE VTE ≥4, prior VTE, high-dose dex, doxorubicin, immobility)
- **Infection: acyclovir/valacyclovir** with PIs, anti-CD38, CAR-T, bispecifics. **Levofloxacin 500 mg ×12 wk** at diagnosis (TEAMM: fever or death HR 0.66). **PJP prophylaxis** with bispecifics, CAR-T, high-dose dex
- **Bispecifics / CAR-T: IVIG 0.4 g/kg q4wk if IgG <400 mg/dL** (hypogammaglobulinemia is near-universal with BCMA agents). Vaccinate: inactivated flu, pneumococcal conjugate, COVID, RSV, **recombinant zoster**; **no live vaccines**; re-vaccinate after ASCT

MYELOMA EMERGENCIES

- **Renal failure / cast nephropathy** (very high involved FLC): **start a bortezomib-based regimen immediately** (e.g. D-VCd / VRd with renal dosing), high-dose dex, hydrate, treat Ca; **stop NSAIDs, avoid IV contrast**. **Plasma exchange and high-cutoff dialysis are not routine** (negative trials). Biopsy if albuminuria → AL, MIDD
- **Hypercalcemia:** IV saline, zoledronic acid (or **denosumab** if renal failure), calcitonin for the first 48 h, steroids + anti-myeloma therapy
- **Cord compression: dexamethasone 10 mg IV now + MRI whole spine** → RT or surgery (instability) · **Hyperviscosity** (IgA, IgG3): **plasmapheresis** · Anemia: treat the myeloma; ESA if Hb <10 despite therapy (VTE risk with IMiDs)

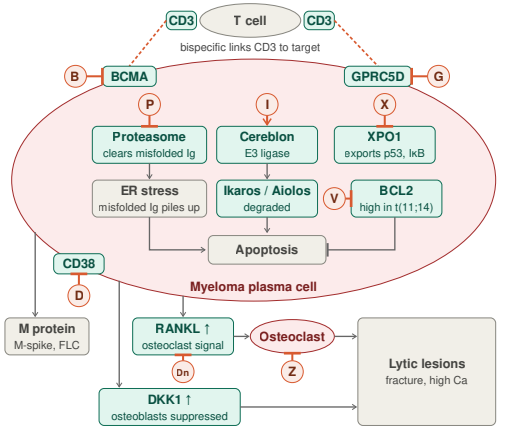
OTHER AGENTS

- **Venetoclax: t(11;14) only, off-label.** BELLINI (Vd ± ven): PFS gain **but excess deaths** (infections), driven by non-t(11;14) — **never outside t(11;14)**
- **Selinexor** (XPO1): SVd weekly — thrombocytopenia 39% grade 3–4, nausea, **hyponatremia**, fatigue; give with 5-HT3 + olanzapine prophylaxis
- **Carfilzomib:** HTN, HF, cardiac events — **baseline echo and BP control**; prefer bortezomib if significant cardiac disease

AGENT-SPECIFIC CONSIDERATIONS

- **Anti-CD38 → blood bank:** dara/isa bind CD38 on reagent RBCs → **panreactive indirect antiglobulin test** for up to 6 mo. Send **type & screen + phenotype/genotype before dose 1**; lab uses **DTT-treated cells** (DTT destroys Kell → give **K-negative units**). ABO typing is unaffected — tell the blood bank
- **Anti-CD38 → SPEP/IFE:** dara (IgGκ) shows up as a small **IgG-κ band** that mimics residual disease — use a DIRA / mass-spec assay before calling non-CR
- **CRS / ICANS:** step-up dosing, **48-h admission** (or monitored outpatient) after step-ups; tocilizumab for grade ≥2 CRS, dex for ICANS; **restart step-up if doses are missed**
- **Cilta-cel: delayed movement/neurocognitive toxicity (parkinsonism)** weeks–months out, cranial-nerve (VII) palsy; **boxed warning for secondary T-cell malignancies** (all CAR-T, Apr 2024), MDS/AML
- **Talquetamab: dysgeusia, dry mouth, dysphagia, weight loss**, skin peeling, **nail loss** — common and dose-limiting; dose spacing helps
- **Belantamab: corneal keratopathy** (ocular AEs 79–89% in DREAMM-7/8) — blurred vision, dry eye; **eye exam at baseline and before doses**; hold/delay by exam findings; REMS

PATHOPHYSIOLOGY



B BCMA: CAR-T cilta-cel, ide-cel · ADC belantamab mafodotin
BCMA×CD3 bispecifics: teclistamab, elranatamab, linvoseltamab
G talquetamab: GPRC5D×CD3
D daratumumab, isatuximab: anti-CD38
P bortezomib, carfilzomib: proteasome inhibitors
X selinexor: XPO1 inhibitor
I lenalidomide, pomalidomide, iberdomide: cereblon modulators
V venetoclax: BCL2 inhibitor, t(11;14)
Dn denosumab: anti-RANKL
Z zoledronic acid: bisphosphonate, osteoclast inhibitor

BCMA is often lost after BCMA-directed therapy: switch target (GPRC5D) next.